

The theory

The error representation and the bound, with proofs

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Goal-Oriented DFR series | Part 3

What we will establish

Everything is elementary: a few lines of functional analysis, no regularity assumed beyond well-posedness.

1. The **setting**: trial and test spaces, the bilinear form, the two hypotheses (Banach–Necas–Babuska).
2. The **DFR error-loss equivalence**, derived.
3. The **exact error representation** for the Qol (a proposition, with proof).
4. The **main bound**: the Qol error is controlled by the goal loss plus a computable remainder (a theorem, with proof).
5. Two **remarks**: the second-order structure, and how standard DFR is recovered.

Section 1

The setting

Spaces, form, and the primal problem

Let U (trial) and V (test) be Hilbert spaces, $a : U \times V \rightarrow \mathbb{R}$ a bilinear form, and $\ell \in V^*$.

Primal problem

Find $u^* \in U$ with $a(u^*, v) = \ell(v) \quad \forall v \in V$.

For $w \in U$, the **residual functional** $R(w) \in V^*$ is

$$R(w)(v) = \ell(v) - a(w, v).$$

Then $R(u^*) = 0$, and subtracting the primal problem,

$$R(w)(v) = a(u^* - w, v) \quad \forall v \in V$$

the identity every proof below leans on.

The two hypotheses

(A1) Boundedness

$$|a(w, v)| \leq M \|w\|_U \|v\|_V \text{ for all } w \in U, v \in V.$$

(A2) Inf-sup stability

There is $\gamma > 0$ with

$$\inf_{w \neq 0} \sup_{v \neq 0} \frac{a(w, v)}{\|w\|_U \|v\|_V} \geq \gamma \quad \text{and} \quad \inf_{v \neq 0} \sup_{w \neq 0} \frac{a(w, v)}{\|w\|_U \|v\|_V} \geq \gamma.$$

- These are the Banach–Necas–Babuska (BNB) hypotheses; they make the primal problem and its adjoint well posed.
- For the model problem (Poisson, $U = V = H_0^1$, energy norm) they hold with $\gamma = M = 1$.

The DFR error-loss equivalence

Theorem (Error-loss equivalence)

Under (A1)–(A2), for every $w \in U$,

$$\frac{1}{M} \|R(w)\|_{V^*} \leq \|u^* - w\|_U \leq \frac{1}{\gamma} \|R(w)\|_{V^*}.$$

Proof.

From the boxed identity, $\|R(w)\|_{V^*} = \sup_{v \neq 0} \frac{a(u^* - w, v)}{\|v\|_V}$. The **upper** bound on $\|R(w)\|_{V^*}$ is (A1) with $w \mapsto u^* - w$, giving $\|R(w)\|_{V^*} \leq M \|u^* - w\|_U$. The **lower** bound is the first inf-sup of (A2) applied to $u^* - w$: $\gamma \|u^* - w\|_U \leq \sup_v a(u^* - w, v) / \|v\|_V = \|R(w)\|_{V^*}$. Rearranging gives both inequalities. $\square$

Section 2

The quantity of interest and its adjoint

The QoI and the adjoint problem

Let the quantity of interest be a bounded linear functional $J \in U^*$.

Adjoint problem

Find $z^* \in V$ with $a(w, z^*) = J(w) \quad \forall w \in U$.

Well posed by (A1) and the *second* inf-sup of (A2). For $z \in V$, the **adjoint residual** $R^*(z) \in U^*$ is

$$R^*(z)(w) = J(w) - a(w, z).$$

Then $R^*(z^*) = 0$, and subtracting the adjoint problem,

$$R^*(z)(w) = a(w, z^* - z) \quad \forall w \in U.$$

Takeaway: Same structure as the primal, with the roles of the two arguments of a swapped.

Section 3

The error representation

Exact error representation

Proposition

For u^* solving the primal and z^* the adjoint, and any $u_h \in U$, $z_h \in V$,

$$J(u^*) - J(u_h) = R(u_h)(z^*) = R(u_h)(z_h) + R(u_h)(z^* - z_h).$$

Proof.

Take $w = u^* - u_h$ in the adjoint problem, then use the primal identity:

$$J(u^*) - J(u_h) = J(u^* - u_h) = a(u^* - u_h, z^*) = R(u_h)(z^*).$$

The second equality is the split $z^* = z_h + (z^* - z_h)$ and linearity of $R(u_h)$. □

The first term is computable: it is the goal loss $\mathcal{L}_{\text{QoI}} = |R(u_h)(z_h)|$. The second involves the unknown z^* and we bound it next.

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Section 4

The main bound

Theorem

Under (A1)–(A2), for every $u_h \in U$ and $z_h \in V$,

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{1}{\gamma} \|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}.$$

- All three terms on the right are exactly what training minimizes.
- The remainder is a **product** of the two residuals: second order when both are small.
- **One-sided**: a reliability bound, not an equivalence. We prove it on the next two slides.

Proof, part 1: split and bound by the dual norm

By the error representation and the triangle inequality,

$$|J(u^*) - J(u_h)| \leq \underbrace{|R(u_h)(z_h)|}_{= \mathcal{L}_{\text{QoI}}(u_h, z_h)} + |R(u_h)(z^* - z_h)|.$$

For the remainder, the definition of the dual norm gives

$$|R(u_h)(z^* - z_h)| \leq \|R(u_h)\|_{V^*} \|z^* - z_h\|_V.$$

It remains to control $\|z^* - z_h\|_V$ by the *adjoint* residual. That is where the second inf-sup enters.

Proof, part 2: inf-sup on the adjoint error

Apply the second inf-sup of (A2) to $z^* - z_h \in V$, then the boxed adjoint identity:

$$\gamma \|z^* - z_h\|_V \leq \sup_{w \neq 0} \frac{a(w, z^* - z_h)}{\|w\|_U} = \sup_{w \neq 0} \frac{R^*(z_h)(w)}{\|w\|_U} = \|R^*(z_h)\|_{U^*}.$$

Chaining the three displays,

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{1}{\gamma} \|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}. \quad \square$$

Takeaway: The bound is fully a posteriori: every term is computed from the two trained networks.

A corollary: adjoint trained to tolerance

Corollary

If $\|R^*(z_h)\|_{U^*} \leq \varepsilon$, then

$$|J(u^*) - J(u_h)| \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{\varepsilon}{\gamma} \|R(u_h)\|_{V^*} \leq \mathcal{L}_{\text{QoI}}(u_h, z_h) + \frac{M\varepsilon}{\gamma} \|u^* - u_h\|_U.$$

- The last step uses the *left* inequality of the DFR equivalence, $\|R(u_h)\|_{V^*} \leq M \|u^* - u_h\|_U$.
- Once the adjoint is accurate, the QoI error is the goal loss plus a term that scales with the primal energy error.

Section 5

Two remarks

Remark: second-order structure, and what it does not say

If both networks reach energy accuracy $O(\delta)$, the remainder is $O(\delta^2)$: driving the goal loss to zero leaves a QoI error quadratic in δ . But:

1. It **presupposes δ small**, i.e. both networks well resolved. In a resolution-limited regime, δ is not small and the bound is weak exactly where the method is wanted.
2. There is **no Galerkin orthogonality** here. The goal loss enforces one scalar condition, not orthogonality against a whole subspace.
3. So $\mathcal{L}_{\text{QoI}}$ is **not an estimator**: training drives it to zero, and the bound is carried by the remainder. Empirically $\mathcal{L}_{\text{QoI}} \approx 10^{-10}$ while the error $\approx 10^{-5}$.

Remark: standard DFR is the worst case over goals

Take the supremum of the error representation over all unit goals. Since

$$\sup_{\|J\|_{U^*}=1} J(u^* - u_h) = \|u^* - u_h\|_U,$$

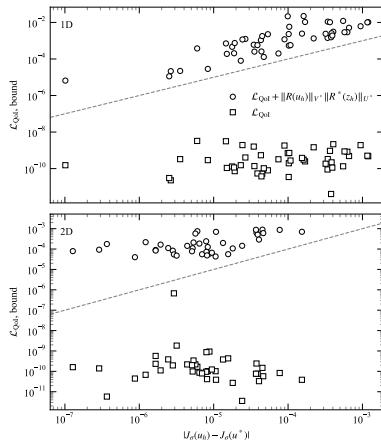
the worst case over goals reproduces the DFR equivalence.

- Standard DFR controls *every* quantity of interest at once.
- Goal-oriented DFR controls a *single* prescribed goal.
- That is why the latter can be cheaper for a fixed target accuracy: it does not pay for the goals you do not care about.

Section 6

The bound, empirically

The bound, verified



- Circles (the bound) sit **above** the diagonal on all 168 runs, at about $15\times$ the true error. Never optimistic, despite mode truncation and quadrature error.
- Squares (the goal loss alone) sit far **below**: it cannot serve as an estimator, just as the remark warned.